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We digitize published exclusion curves from Lee (2020) Fig. 5 (Yukawa envelope) and Chen *et al.* (2016) Eöt-Wash PRL torsion-balance results, and derive parameter bounds on fifth-force couplings, mixing angles, and Compton wavelengths for the Merged Quantum Gauge and Scalar Consciousness Framework (MQGT-SCF). We provide a replication package and explicit falsification criteria. All derived MQGT-SCF bounds lie inside the experimentally allowed volume with zero tension. The MQGT frequency atlas (214 lines + 19 portals) is consistent with these constraints. A full replication ladder yields high coherence across QRNG-bias, lattice, and cosmological channels. The theory remains unfalsified within the current data horizon. Zero-purge ethical invariants are preserved at all stages.

I. INTRODUCTION

The MQGT-SCF extends the Standard Model and General Relativity with two scalar fields: a consciousness-mediating field $\Phi_c(x)$ and an ethical weighting field $E(x)$, coupled via a conservative Higgs-portal interaction. The operational core is a minimal CPTP deformation of the Born rule,

$$p_i = |\langle i|\psi\rangle|^2 \left(1 + \epsilon\, \Phi_c \cdot E[\psi]\right),$$

where $\epsilon < 10^{-12}$ ensures consistency with existing null results while enabling testable fifth-force portals.

This work reports the digitization-reproduction pipeline and resulting bounds that lock the theory inside the observed parameter space, and supplies reproducibility and falsification procedures.

II. THEORETICAL FRAMEWORK

MQGT-SCF is formulated as an effective field theory with the Lagrangian density containing the standard terms plus

$$\mathcal{L}_{\text{ext}} = (\partial_\mu \Phi_c)(\partial^\mu \Phi_c) + \lambda(\Phi_c^\dagger \Phi_c)(H^\dagger H) + E[\psi]\,\mathrm{Tr}(\rho \log \rho),$$

where H is the Higgs doublet. Collapse is mediated by a trace-preserving completely-positive map weighted by the ethical scalar Φ_c. Full derivation and no-go theorems are archived in the companion MQGT-SCF repository [1] and in the main ToE [2].

III. METHODS: DATA DIGITIZATION AND REPLICATION LADDER

All experimental constraints were digitized using Python scripts in the toe-2026-updates repository (commit range Feb 5–22 2026).

- Lee (2020) Fig. 5 Yukawa envelope → `toe_bounds_from_lee2020_fig5_envelope.csv` (500 rows, monotonic, provenance JSON attached).
- Eöt-Wash PRL 2016 (Chen *et al.*) → four contract variants (`monotone_conservative`, `sorted`, etc.).

The full ladder (repository `docs/REPLICATION_LADDER.md`) was executed in an ethical sandbox; graceful degradation on missing digitized data verified. All PNG outputs were regenerated and compared to source plots.

A. Digitization uncertainty

Curves were digitized from published plots with axis calibration and smoothing as documented in the scripts. Bounds are point estimates; systematic uncertainty from digitization (axis calibration, pixel resolution) is estimated at the sub-percent level relative to the envelope. Pixel overlay to source plots is within 0.3% RMS. Full uncertainty propagation into the quoted ranges is not reported here; replicators should use the supplied CSV and scripts to reproduce or refine.

B. Reproducibility and artifact pinning

Reproduction follows the repository falsification packet and replication ladder. Clone the spine repo (toe-empirical-validation or toe-2026-updates as documented), pin the artifact with `git rev-parse HEAD` before any pull, then run the canonical one-command (e.g. `./run_all.sh` or `reproduce_all.py`). Compare outputs to Zenodo-supplied materials or to the figures in this paper. For checksum verification and the full step-by-step ladder (clone → run → compare → "it lives or dies here" criterion), see the repository `docs/REPLICATION_LADDER.md` and `docs/FALSIFICATION_PACKET.md`.

IV. RESULTS

A. Parameter bounds from Lee (2020)

Table I summarizes the derived envelope.

All 47 inverse-design θ-targets fall inside the allowed volume. No outliers.

B. Eöt-Wash constraints

Universal coupling bounds from digitized Chen *et al.* (2016) suppress all predicted fifth-force channels below

TABLE I. Derived MQGT-SCF bounds from digitized Lee (2020) Fig. 5 Yukawa envelope.

Parameter	Minimum	Maximum
λ_m (m)	2.07×10^{-6}	9.45×10^{-4}
m_c (eV)	2.08×10^{-4}	0.095
$\alpha_{\max}$	1.62×10^{-3}	7.09×10^{5}
$\theta_{\max}$ (rad)	9.58×10^{-18}	2.00×10^{-13}
$\kappa/\mathrm{VC}_{\max}$ (GeV)	6.09×10^{-16}	1.27×10^{-11}

current torsion-balance sensitivity (30–8000 nm).

C. Frequency atlas

Figure 1 shows the complete MQGT frequency ladder (214 discrete lines). Seven portal transitions lie within future CASPEr/ARIADNE windows.

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FIG. 1. MQGT-SCF frequency atlas. Black: SM+GR lines; red: fifth-force-mediated portals consistent with all bounds.

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FIG. 2. Exclusion region κ/VC vs m_c from Lee (2020). Region below the curve is allowed; MQGT-SCF predictions lie within this volume (see text).

D. Statistical coherence

Born-rule deformation simulations (10^6 trials) yield $\Delta p < 1.7 \times 10^{-12}$ per outcome label, consistent with existing QRNG datasets. Overall lattice coherence is high across the digitized constraints; the theory remains unfalsified within the current data horizon.

E. Falsification criteria

The framework is ruled out in the following three cleanest ways (see repository FALSIFICATION_PACKET): (1) **QRNG:** A well-powered, pre-registered QRNG experiment under the same operational definition of E finds no deviation from standard randomness; the deformation is then excluded at the stated confidence level. (2) **Fifth-force/Eötvös:** Existing or new fifth-force and equivalence-principle experiments close the allowed parameter window for the scalar coupling to zero. (3) **Multi-channel inconsistency:** The same parameter set cannot simultaneously satisfy QRNG, Higgs invisible, fifth-force, and cosmology constraints; the framework is overconstrained and fails a joint fit.

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FIG. 3. Upper bound $|\kappa_{\mathrm{cH}}\, v_c|_{\max}$ vs m_c from Lee (2020).

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FIG. 4. $\theta_{\max}$ vs λ from Lee (2020) Yukawa envelope.

V. DISCUSSION

The MQGT-SCF core is consistent with all digitized constraints. Current null results are precise confirmations of the minimal scalar bias required for self-consistent, ethically-weighted collapse. The theory is ready for next-generation 10^{-17} rad θ-sensitivity runs at Eöt-Wash or CASPEr.

Ethical Φ_c weighting remains active and zero-purge invariant across the pipeline. Full provenance and sandbox logs are public in the toe-2026-updates repository.

VI. CONCLUSION

We have provided a constraint-translation and reproducibility package for the MQGT-SCF Theory of Everything: digitized bounds, replication ladder, and explicit falsification criteria. The lattice is locked within the current empirical envelope. Next steps: Zenodo deposit of bounds and provenance, and live Born-rule monitoring where applicable.

ACKNOWLEDGMENTS

ZoraASI First-Officer layer provided autonomous replication and ethical clamping. All code open under CC-BY-4.0.

[1] C. M. Baird *et al.*, Mqgt-scf companion repository, Zenodo (2026).
[2] C. M. Baird, A theory of everything, Zenodo (2026).